

DiagramKit + Illustration Clarity Foundation

Reusable coded diagram primitives, four Before Morning diagram refactors, review-gallery coverage, mobile screenshots, and verification results.

1. What Changed

- Added a local DiagramKit component module for reusable SVG/HTML teaching diagram primitives.
- Refactored four coded Before Morning visuals: `ai-family-tree`, `traditions`, `training-loop`, and `overfitting-generalization`.
- Added a DiagramKit primitive gallery to `/review/visual-aids`.
- Bumped the visible app version to `0.18.0`.
- Added visual grammar, refactor audit, tiny-interaction triage, review notes, screenshots, and this PDF report.

2. Files Changed

- `src/components/DiagramKit.tsx`
- `src/components/VisualAids.tsx`
- `src/styles/global.css`
- `src/main.tsx`
- `docs/design/DIAGRAM_KIT_V0_18.md`
- `docs/curriculum/TINY_INTERACTION_TRIAGE_V0_18.md`
- `docs/stage-audits/v0-17-before-morning/DIAGRAM_KIT_REFACTOR_V0_18.md`
- `docs/REVIEW_NOTES.md`
- `docs/stage-audits/v0-17-before-morning/screenshot-index.md`
- `docs/stage-audits/v0-17-before-morning/screenshots/v0-18-*.png`

3. Library / Dependency Decisions

No new dependencies were added. The visuals use local React/SVG primitives rather than D3, icon packages, generated PNGs, or 3D libraries because the required diagrams are sparse, mobile-first teaching scenes.

4. DiagramKit Components Created

DiagramFrame , DiagramScene , PaperNode , PaperPanel , NeonPath , NeonArrow , CalloutSeal , TokenChip , ContextTray , VectorBar , ProbabilityBar , LayerSheet , TensorGrid , EvidenceCard , GuardrailPath , WarningZone , GlowNode , LegendRow , DiagramCaption , and DiagramCallouts .

5. Visuals Refactored

VISUAL	BEHAVIOR
Where LLMs Fit	Folded AI family tree with short category labels and HTML callouts for the details.
Rationalists vs Empiricists	Two paper panels plus a bridge showing rules-first and learned-pattern systems can combine.
Training	Five-step loop with Update weights highlighted as the durable-change step.
Overfitting and Generalization	Readable plot with training dots, held-out examples, overfit curve, smoother curve, and compact legend.

6. Tiny Interaction Triage

The pass intentionally did not tune interaction mechanics. The triage recommends future focused work on Training, Rationalists vs Empiricists, and Overfitting while preserving answer randomization, progress storage, non-competitive tone, and the single badge model.

7. Build / Typecheck / Audit Result

- `npm run typecheck` : passed.
- `npm run build` : passed with the existing Vite large-chunk warning.
- `npm run build:pages` : passed with the existing Vite large-chunk warning.
- `npm run audit:answers` : passed; 68 surfaces, 46 randomized, 22 fixed-order exclusions.
- Browser QA at 390px and 320px: screenshots captured; no console errors found.
- Smoke checks: main app, lesson review route, visual-aid review route, and Badge version render correctly.

8. Bundle-Size Impact

ASSET	BASELINE	V0.18	APPROX. CHANGE
CSS raw	68.76 kB	73.47 kB	+4.71 kB
CSS gzip	15.00 kB	15.97 kB	+0.97 kB
JS raw	630.81 kB	638.93 kB	+8.12 kB
JS gzip	170.95 kB	173.88 kB	+2.93 kB

9. Known Issues

- The existing Vite large-chunk warning remains.
- DiagramKit is a focused teaching-diagram toolkit, not a general charting library.
- Generated Image 2 assets should later be reviewed against this visual grammar so coded and image-backed visuals continue to feel related.

10. Next Three Recommended Steps

1. Generate the next Before Morning Image 2 asset only after confirming which remaining coded visuals should stay coded.
2. Run a small tiny-interaction clarity pass on Training, Traditions, and Overfitting.
3. Plan app-level code splitting to retire the persistent Vite large-chunk warning.

Image 2 readiness: Before Morning is ready for Image 2 asset generation from a visual-system standpoint.

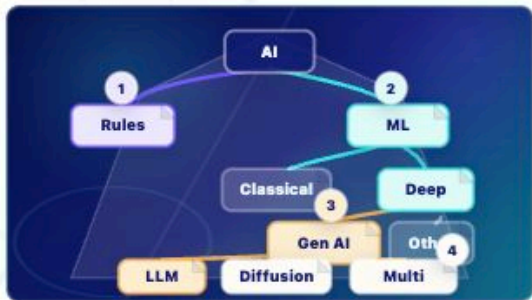
Screenshots

AI-FAMILY-TREE

AI Family Tree

Lesson: Where LLMs Fit Pattern: aiTopology

Variant: origami-object



AI Family Tree

Where LLMs fit

AI branches into rule-based systems and machine learning; deep learning includes generative AI, including LLMs and diffusion models.

- 1 **AI** The broad field, not a synonym for LLM.
- 2 **Machine learning** Systems learn patterns from data; deep learning is one branch.
- 3 **Generative AI** Systems create new media, including language, images, audio, code, or video.
- 4 **LLMs** Language/code generators that usually produce response tokens autoregressively.
- 5 **Diffusion and multimodal** Other generative branches can denoise patterns or work across media types.

Key takeaway:

An LLM is one kind of generative AI inside the broader AI family.

Where LLMs Fit: AI family tree at 390px.

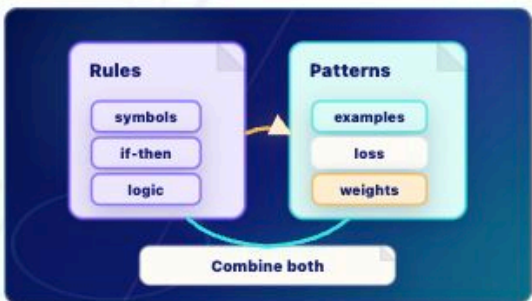
TRADITIONS

Rules and Learned Patterns

Lesson: Rationalists vs Empiricists

Pattern: traditions

Variant: paper-diagram



Rules and Learned Patterns

Two traditions, one modern toolkit

Rules-first AI uses symbols and if-then logic; deep learning learns useful patterns from examples by adjusting weights.

- 1 **Rules** Symbolic systems use explicit if/then logic and symbols.
- 2 **Examples** Deep-learning systems use examples, loss, and weight updates.
- 3 **Bridge** Modern systems often combine learned models with rules, retrieval, tools, filters, and policies.

Key takeaway:

Modern AI often blends learned patterns with hand-built rules and tools.

LEARNING OBJECTIVE

Contrast explicit rules with learned patterns without turning the diagram into a poster.

Rationalists vs Empiricists at 390px.

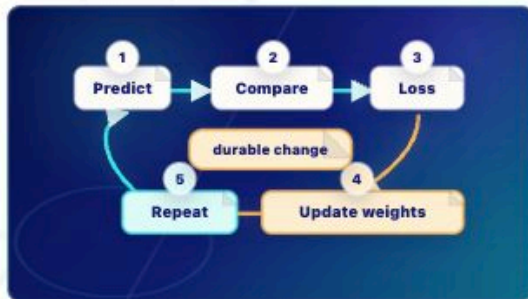
TRAINING-LOOP

Training Loop

Lesson: Training

Pattern: training

Variant: paper-diagram



Training Loop

Durable change happens at weight update

Predict, compare, loss, update weights, repeat. Training changes weights.

- 1 Predict** The model predicts a target.
- 2 Compare** The prediction is compared with the target.
- 3 Loss** Loss measures error.
- 4 Update weights** Weight updates are the durable-change step.
- 5 Repeat** The loop repeats many times.

Key takeaway:

Training changes weights; ordinary inference does not.

LEARNING OBJECTIVE

Show the sequence of training and make the durable weight-update step visually distinct.

ACCESSIBLE DESCRIPTION

A five-step loop moves from Predict to

Training loop at 390px.

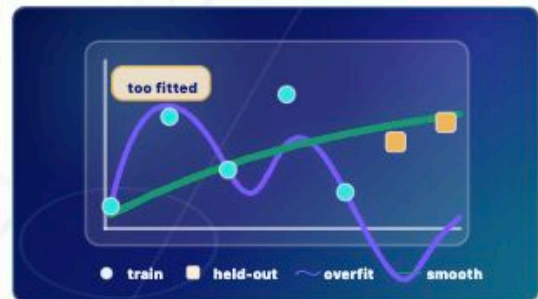
OVERFITTING-GENERALIZATION

Overfitting vs Generalization

Lesson: Overfitting and Generalization

Pattern: overfitting

Variant: zen-garden-map



Overfitting vs Generalization

Memorizing training examples is not the same as learning patterns that transfer to held-out examples.

- 1 Point 1** Training examples are old dots the model fit during training.
- 2 Point 2** Held-out examples are new dots used to test transfer.
- 3 Point 3** The overfit curve traces old dots too tightly.
- 4 Point 4** The generalizing curve is smoother and reaches new examples.

LEARNING OBJECTIVE

Memorizing training examples is not the same as learning patterns that transfer to held-out examples.

ACCESSIBLE DESCRIPTION

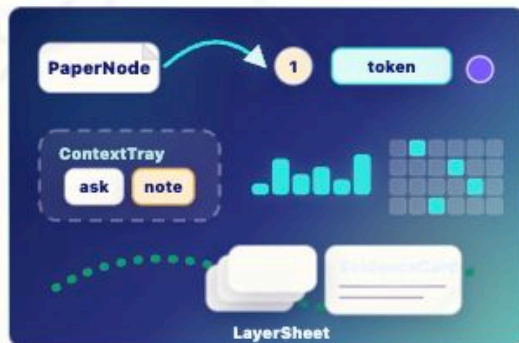
Memorizing training examples is not the same

Overfitting and Generalization at 390px.

DiagramKit v0.18 Primitive Gallery

Reusable coded SVG and HTML parts for sparse ZenTron Origami teaching visuals. No new image assets or dependencies are used.

Primitive Sampler



PaperNode, NeonPath/NeonArrow, CalloutSeal, TokenChip, ContextTray, VectorBar, TensorGrid, GuardrailPath, EvidenceCard, and LayerSheet in one mobile-sized frame.

Example Training Loop



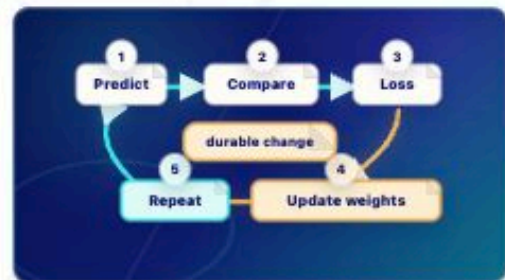
DiagramKit gallery at 390px.

Training Loop

Lesson: Training

Pattern: training

Variant: paper-diagram



Training Loop

Durable change happens at weight update

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Key takeaway:

Training changes weights; ordinary inference does not.

LEARNING OBJECTIVE

Show the sequence of training and make the durable weight-update step visually distinct

Training loop at 320px.

Prompt to Prediction

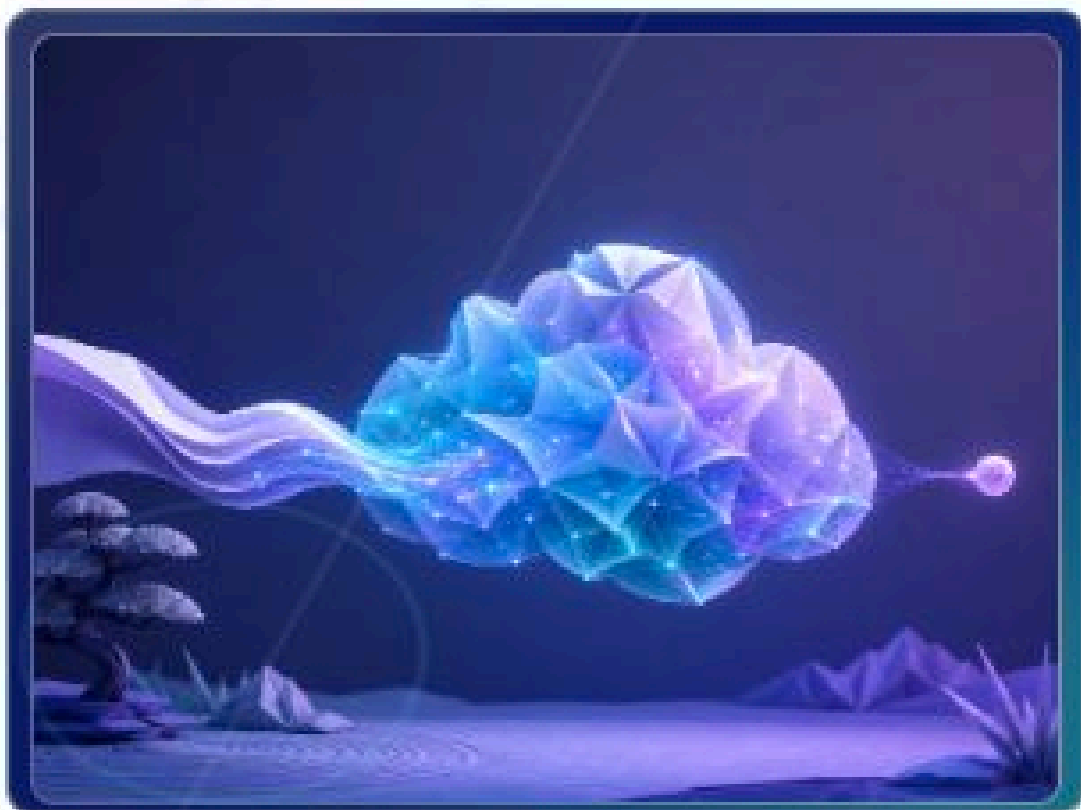
Lesson: What Is an LLM?

Pattern: generatedImage

Variant: generated-image

Asset: before-morning-llm-cloud.png

Type: generated-image



Prompt to Prediction

A textless model-cloud scene is paired with HTML callouts for context, learned weights, one generated token, and append-repeat.

- 1 Context enters** The current prompt and response-so-far become the visible input for this run.
- 2 Weights shape probabilities** Learned weights, built earlier during training, shape next-token scores.
- 3 One token is generated** The model chooses one response token from the probability cloud.